CIE AS PHYSICS (9702) • QUESTION PAPER

8.3 Interference

Syllabus Section: 8. Superposition **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (81 Marks)

Question 1 Easy [11 marks]

Part (a) [2 marks]

Define *coherence*.

Part (b) [4 marks]

In an experiment, a monochromatic light source is incident upon a single slit before passing through two slits, as shown in Fig. 1.1.



Fig. 1.1

Bright and dark fringes are produced on the screen.

State

(i) the **two** conditions required for the fringes to be observed.

[2]

(ii) what the bright and dark fringes on the screen represent.

[2]

Part (c) [1 mark]

Give a reason why a single slit is included in the experimental setup in Fig. 1.1.

Part (d) [4 marks]

Fig. 1.2 shows two different waves from sources s_1 and s_2 meeting on the screen at point P.

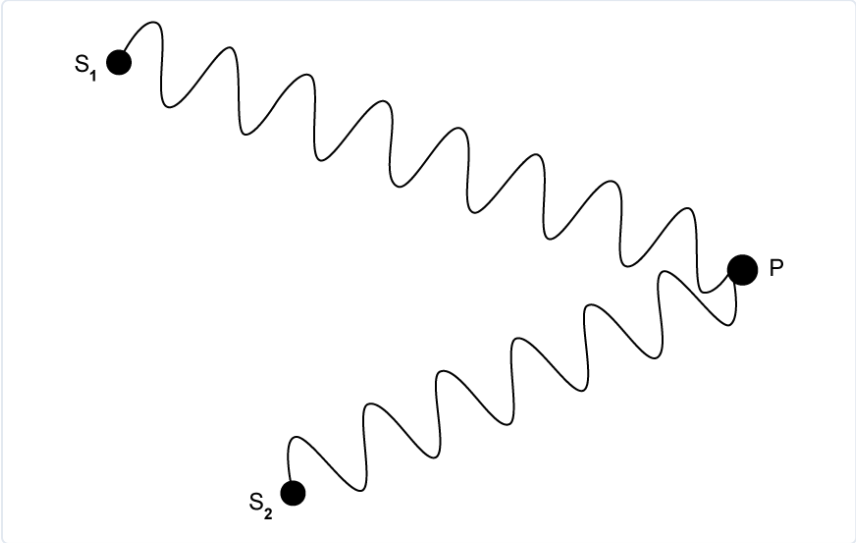


Fig. 1.2

For the waves meeting at point P, determine:

(i) the path difference.

[3]

(ii) whether a bright or dark fringe forms at P.

[1]

Question 2 Easy [8 marks]

Part (a) [1 mark]

State what is meant by the term *monochromatic source*.

.....

.....

Part (b) [1 mark]

Describe the conditions for destructive interference.

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.....

Part (c) [4 marks]

Fig. 1.1 shows the wavefronts emitted from two point sources s_1 and s_2.

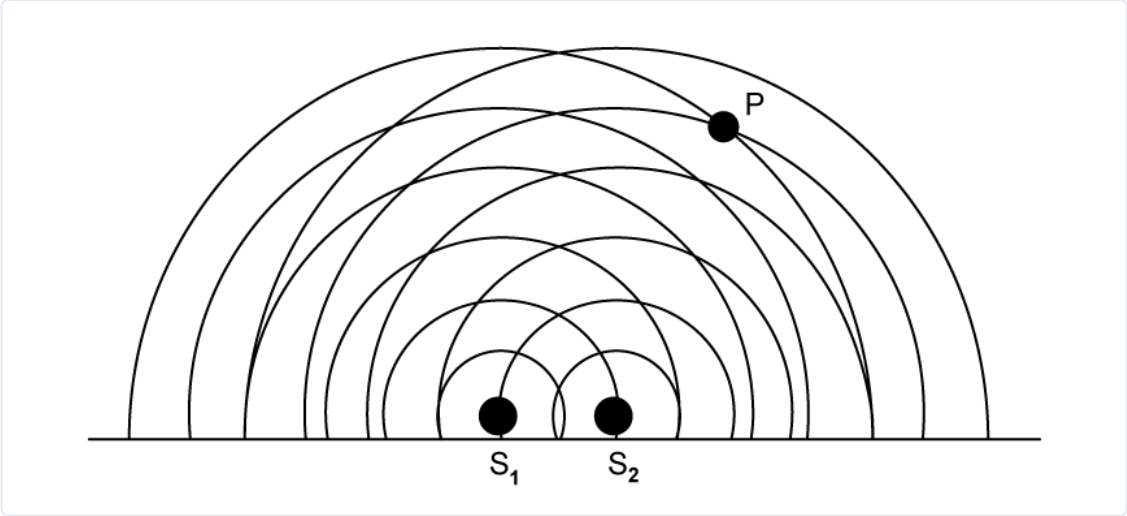


Fig. 1.1

At point P, determine

(i) the path difference between the waves

[3]

(ii) whether constructive or destructive interference occurs.

[1]

.....

.....

Part (d) [2 marks]

Fig. 1.2 shows two sources of coherent light producing a double-slit interference pattern.

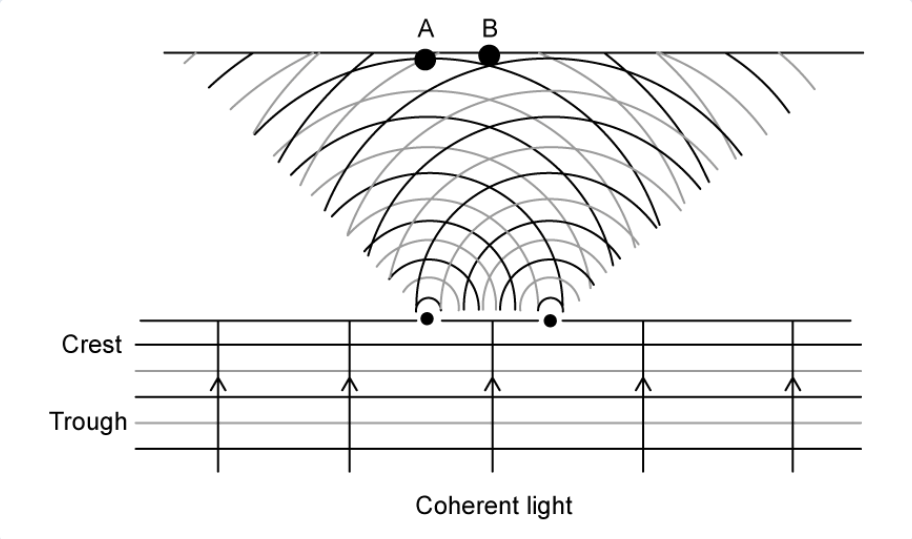


Fig. 1.2

Determine whether constructive or destructive interference occurs at:

(i) point A

[1]

(ii) point B.

[1]

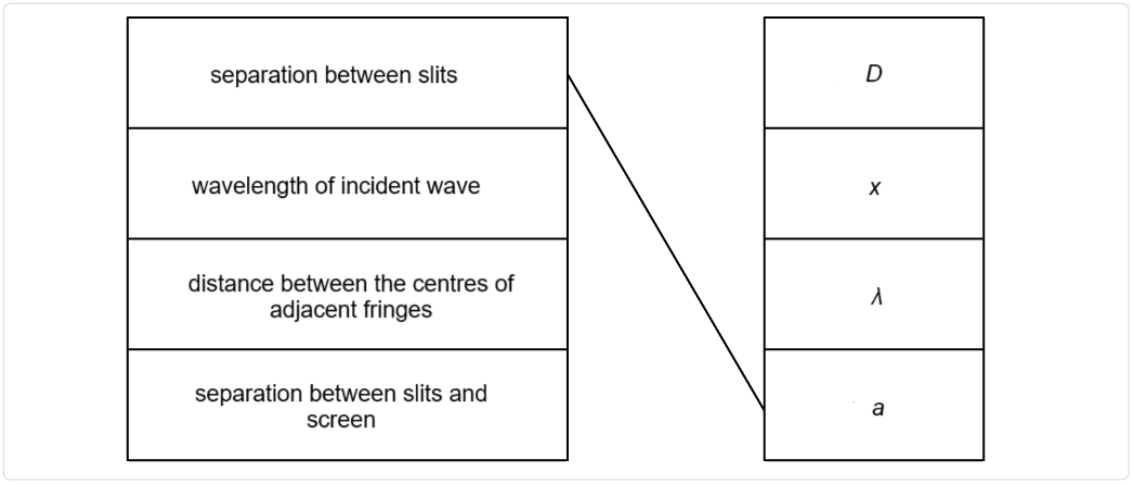
Question 3 Easy [9 marks]

Part (a) [3 marks]

The distance between the bright fringes in a double-slit interference pattern can be determined by the double-slit equation

$$x = \frac{\lambda D}{a}$$

Draw lines to match each quantity to its associated symbol.



Part (b) [2 marks]

Red laser light is used to form a double-slit interference pattern on a screen, as shown in Fig. 1.1.

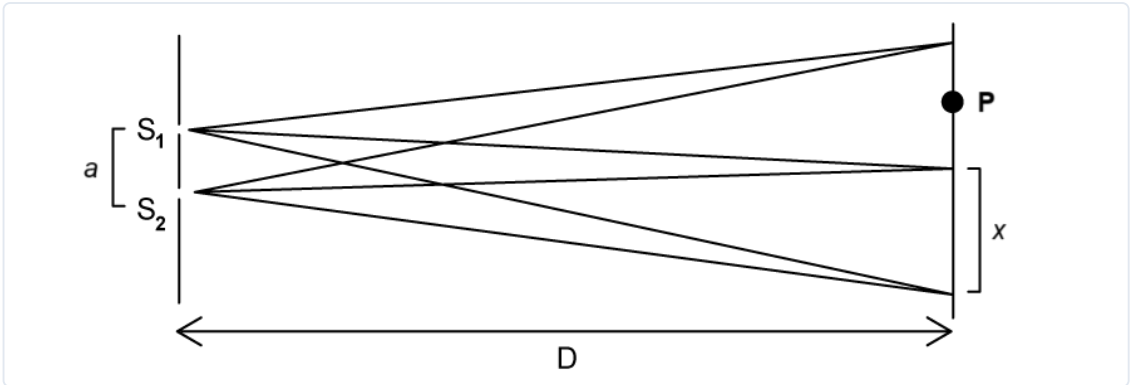


Fig. 1.1

The distance between the bright fringes depends on the wavelength of the incident light.

Describe how the interference pattern would change if blue laser light were used.

Part (c) [2 marks]

When red laser light of wavelength 650 nm is used to form a double-slit interference pattern on the screen, the separation between the slits is 0.2 mm, and the distance between the slits and the screen is 1.2 m.

Calculate the separation between the fringes on the screen.

Part (d) [2 marks]

Explain why there is no bright fringe at point P.

Question 4 Medium [12 marks]

Part (a) [3 marks]

Fig. 1.1 shows an arrangement for observing the interference pattern produced by laser light passing through two narrow slits S_1 and S_2.

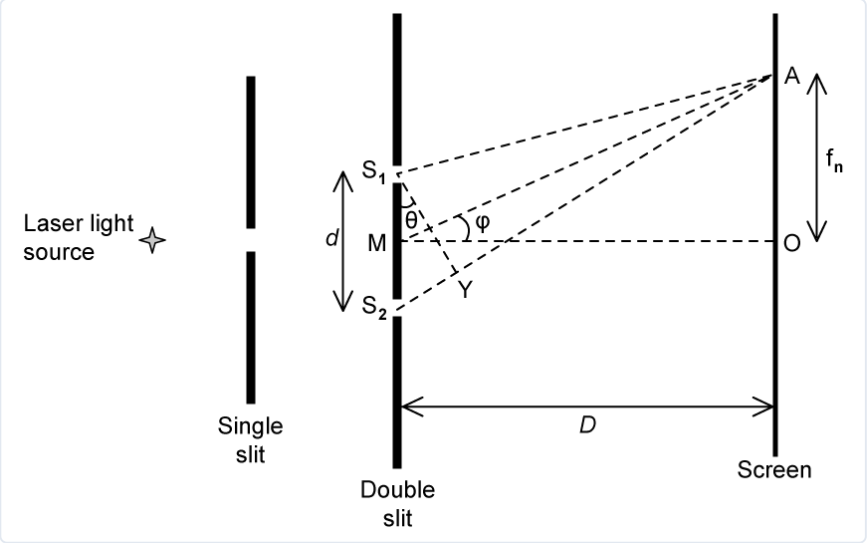


Fig. 1.1

The distance S_1S_2 is d, and the distance between the double slit and the screen is D, where $D \gg d$, so angles θ and ϕ are small.

M is the midpoint of S_1S_2 and it is observed that there is a bright fringe at point A on the screen, a distance f_n from point O on the screen. Light from S_2 travels a distance S_2Y further to point A than light from S_1.

The wavelength of light from the laser is 650 nm and the angular separation of the bright fringes on the screen is 5.00×10^{-4} rad.

Calculate the distance between the two slits.

Part (b) [3 marks]

A bright fringe is observed at A.

(i) Explain the conditions required in the paths of the rays coming from S_1 and S_2 to obtain this bright fringe.

[2]

(ii) State an equation in terms of wavelength for the distance S_2Y.

[1]

Part (c) **[4 marks]**

Deduce expressions for the following angles in the double-slit arrangement shown in part a:

(i) θ in terms of S_2Y and d

[2]

(ii) ϕ in terms of D and f_n

[2]

Part (d) **[2 marks]**

The separation of the slits S_1 and S_2 is 1.30 mm.

The distance MO is 1.40 m.

The distance f_n is the distance of the ninth bright fringe from O and the angle θ is 3.70×10^{-3} radians.

Calculate the wavelength of the laser light.

Question 5 Medium [11 marks]

Part (a) [6 marks]

A student is conducting a series of investigations on diffraction and interference with two slits.

In the first investigation, monochromatic light passes through a double-slit arrangement. The intensity of the fringes varies with distance from the central fringe. This is observed on a screen, as shown in Fig. 1.1.

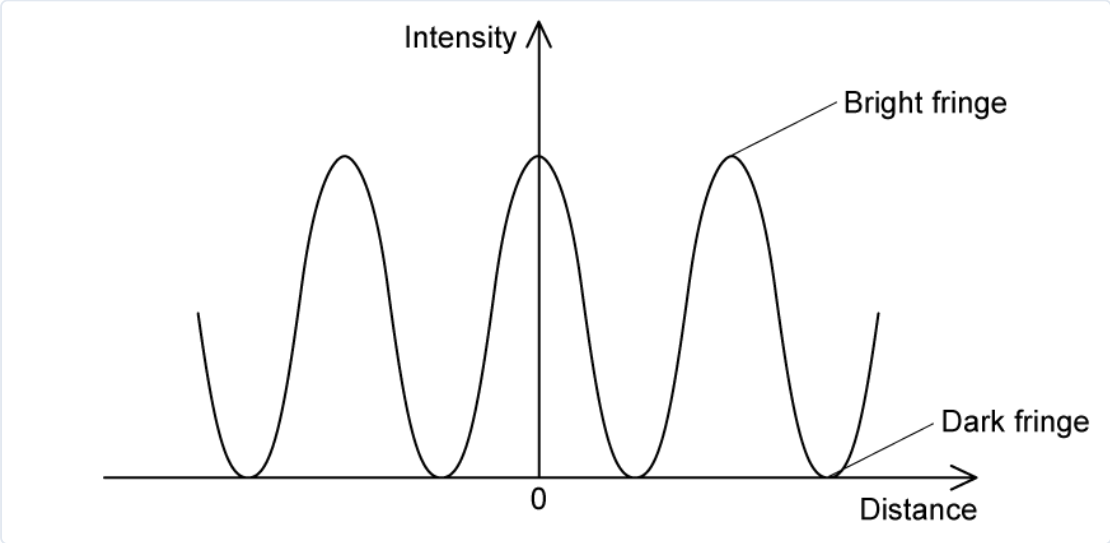


Fig. 1.1

The intensity of the monochromatic light passing through one of the slits is reduced.

Explain the effect of this change on the appearance of

(i) The dark fringes.

[3]

(ii) The bright fringes.

[3]

Part (b) [2 marks]

In another experiment, the monochromatic light is replaced by orange light of wavelength 600 nm. The double-slit has a separation of 0.350 mm and the screen is 6.35 m away.

Calculate the distance between the central and first maximum as seen on the screen.

Part (c) [3 marks]

The light source is now changed to a blue LED of wavelength 450 nm.

Explain the features of the interference pattern that will now be observed on the screen.

Question 6 Medium [8 marks]

Part (a) [3 marks]

Fig. 1.1 shows schematically an arrangement for producing interference fringes using a double slit.

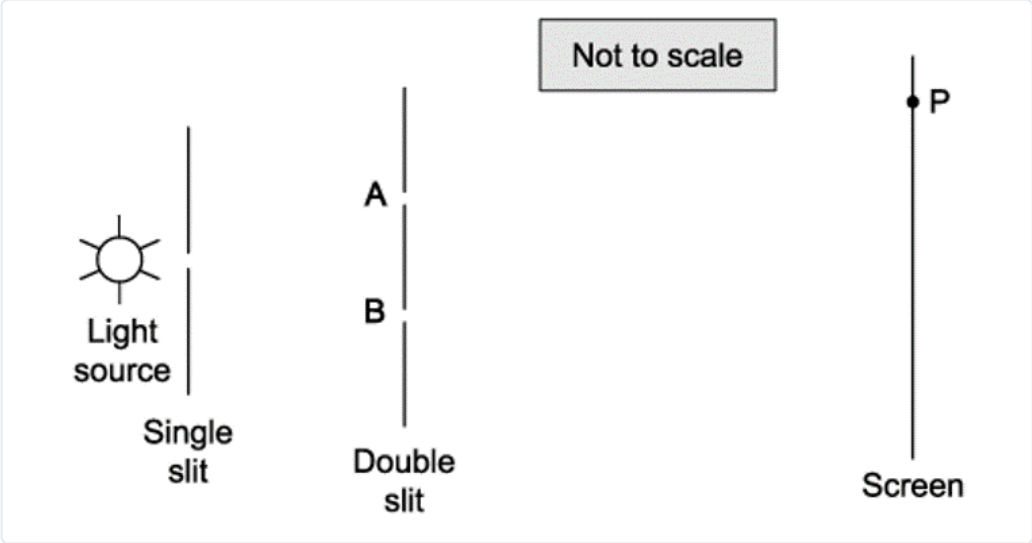


Fig. 1.1

There are two slits at points A and B and a bright fringe (maximum intensity) is observed at the point labelled P.

Explain how the path difference determines that the light intensity at point P is a maximum.

Part (b) [3 marks]

The light source has a wavelength of 45 mm.

The distance from A to P = 1.13 m and from B to P is 1.40 m.

Show that the light intensity at point P is a maximum.

Part (c) [2 marks]

The light source used in Fig. 1.1 is monochromatic but not coherent.

Explain the purpose of the single slit to observe clear interference fringes.

Question 7 Hard [12 marks]

Part (a) [3 marks]

A student designs an experiment to replicate Young's double-slit demonstration.

The student uses a candle as a light source, with a piece of coloured filter paper to produce monochromatic light. They consider additional apparatus required to observe an interference pattern.

Sketch a diagram of the experimental setup the student should use to observe an interference pattern.

Label all apparatus and relevant measurements to be taken.

Part (b) [4 marks]

The student labels the two slits on the double-slit grating slit X and slit Y.

The student then paints over slit X, such that the intensity of light emerging from it is 50% of the intensity emerging from slit Y.

Discuss the effects, if any, this will have on the student's observations.

Part (c) [3 marks]

The student finishes setting up their apparatus and makes a quick note of two separate measurements in a lab book as shown in Fig. 1.1.



Fig. 1.1

They then plot a graph of the intensity of light against the distance from the centre of the screen, represented by the origin as shown in Fig. 1.2.

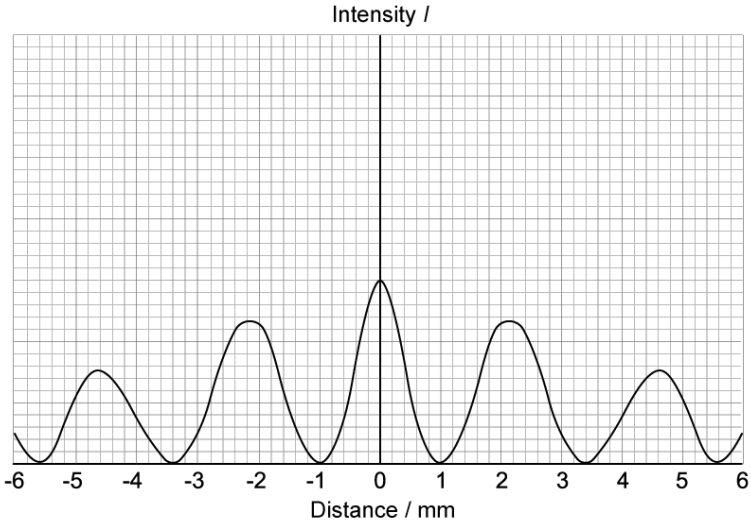


Fig. 1.2

Determine which colour of filter paper the student most likely chose for this experiment, using the information in Fig. 1.1 and Fig. 1.2.

Part (d) [2 marks]

Determine the phase angle between the waves meeting at the point that is 2.8 mm from the centre of the screen.

Question 8 Hard [10 marks]

Part (a) [3 marks]

Two coherent sources, A and B, which are in phase with each other, emit microwaves of wavelength 40.0 mm.

A detector is placed at the point P where it is 0.93 m from A and 1.19 m from B, as shown in Fig. 1.1. The centre axis is normal and a bisector to the straight line joining A and B.

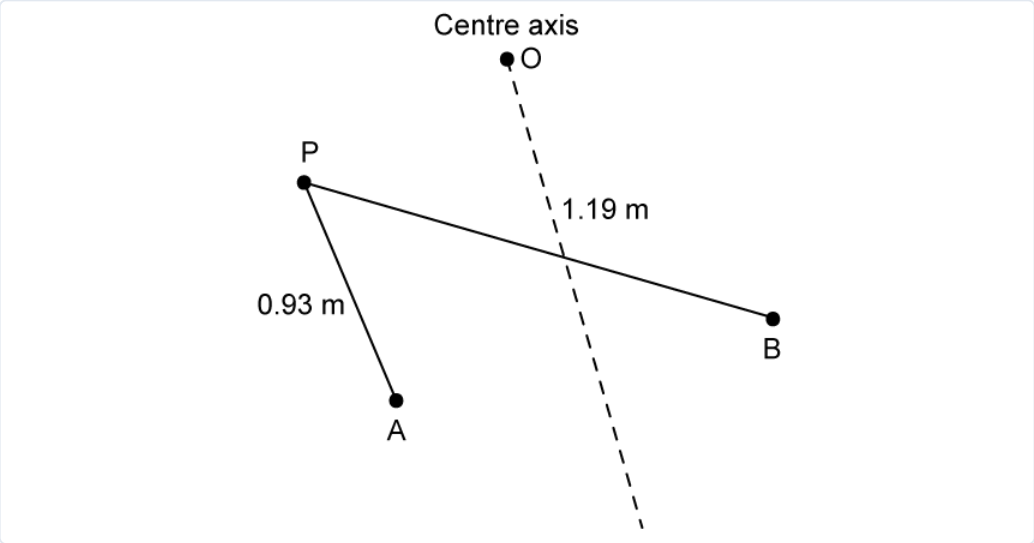


Fig. 1.1

Deduce whether the detected signal at P is a maximum or minimum.

Part (b) [3 marks]

The amplitude of the waves emitted from source B is twice that of the waves emitted from source A.

Calculate the ratio of the intensity at P to the intensity at O.

Part (c) [2 marks]

Determine the total number of maxima and minima detected by the detector as it moves from P to O.

Part (d) [2 marks]

Source B is altered to emit waves that are 180° out of phase with source A.

Describe and explain the change in the appearance of the interference pattern at P.